SSC MTS 2024 question paper with answers

23 October 2024 · Shift 1

75 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 15 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.2

The average of 41 numbers is 62. The average of the first 18 numbers is 65.5, and the average of the last 24 numbers is 61.5. If the 18th number from the beginning is excluded, then what is the average of the remaining numbers?

(A) 61.625
(B) 60.725
(C) 61.325
(D) 60.5

Q.3

A girl spends 76% of her income. If her income increases by 18% and her expenditure increases by 25%, then what is the percentage increase or decrease in her savings? (Correct to one decimal place)

(A) Increase by 3.6%
(B) Increase by 5%
(C) Decrease by 4.2%
(D) Decrease by 3.8%

Q.5

An ice-cream cone with a lid has a total surface area equal to the area of another circle of certain radius. If the radius of the circle is equal to average of the slant height and the radius of the cone, then the ratio of the slant height of the cone to the radius of the circle is:

(A) 2 : 3
(B) 3 : 1
(C) 3 : 2
(D) 1 : 3

Q.9

The difference between successive discounts of 28% followed by 25% and 10% followed by 5% on the marked price of a pen is ₹63. The marked price of the pen is:

(A) ₹200
(B) ₹150
(C) ₹165
(D) ₹331

Q.10

Find the LCM of 120, 200 and 225.

(A) 1200
(B) 2400
(C) 1800
(D) 1735

Q.11

A boat is rowed down along a river 21km in 3 hours and against a river 14km in 7 hours. Find the speed of the boat and the river.

(A) 6.4 km/h and 4.6 km/h
(B) 3.2 km/h and 4.8 km/h
(C) 2.6 km/h and 4.8 km/h
(D) 4.5 km/h and 2.5 km/h

Q.12

A train moving at the same speed passes two bridges of length 900 m and 486 m in 110 seconds and 70 seconds, respectively. The length of the train (in m) is:

(A) 283.5
(B) 253.8
(C) 225.7
(D) 238.5

Q.13

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 4% per year rate of interest for 2 years.

(A) 240
(B) 220
(C) 320
(D) 280

Q.15

A hemispherical bowl with an internal radius of 27 cm is full of liquid. The liquid is to be filled into cylindrical shaped bottles of diameter 12 cm and height 13.5 cm. The number of bottles required to empty the bowl is:

(A) 27
(B) 29
(C) 21
(D) 25

Q.16

The average weight of 11 persons is increased by 4.5 kg when one of them, whose weight is 50.5 kg, is replaced by a new person. The weight of the new person is:

(A) 95.5 kg
(B) 100 kg
(C) 90.5 kg
(D) 98.5 kg

Q.18

The average of five numbers is 62. The average of the first two numbers is 48 and that of the last two numbers is 52. What is the third number?

(A) 110
(B) 100
(C) 95
(D) 105

Q.19

What sum (in ₹) will earn a simple interest of ₹240 in 2 years at 8% per year rate of interest?

(A) 2500
(B) 1500
(C) 750
(D) 3000

Q.20

What should be added to each of the four numbers 5, 19, 7 and 25 to make them proportional?

(A) 2
(B) 1
(C) 3
(D) 4

Reasoning Ability and Problem Solving 13 questions

Q.2

A, B, C, D, E, F, and G are sitting around a circular table, facing away from the centre (but not necessarily in the same order).Only 2 people sit between A and D when counted from the right of D.B and C are both immediate neighbours of G.F sits to the immediate left of E.Only 1 person sits between C and E when counted from the left of C. How many people are sitting between G and D when counted from the right of G?

(A) 1
(B) 3
(C) 2
(D) 4

Q.3

RXPL is related to OUMI in a certain way based on the English alphabetical order. In the same way, TZRN is related to QWOK. To which of the following is PVNJ related, following the same logic?

(A) SMGK
(B) MSGK
(C) MSKG
(D) SMKG

Q.6

4 is related to 17 following a certain logic. Following the same logic, 5 is related to 26. To which of the following is 10 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 101
(B) 99
(C) 121
(D) 103

Q.7

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 69 – 24 ÷ 4 × 6 + 13 = ?

(A) 72
(B) 70
(C) 68
(D) 86

Q.10

In a certain code language, 'have good food' is coded as 'ti ma ka' and 'nutritious food healthy' is coded as 'ka lu pe'. How is 'food' coded in the given language?

(A) ti
(B) ka
(C) lu
(D) ma

Q.12

'4KC2' is related to '6MA4' in a certain way. In the same way, '3MN4' is related to '5OL6'. To which of the following is '5NO3' related following the same logic?

(A) 6MO7
(B) 5OM7
(C) 7PM5
(D) 8PL6

Q.13

What should come in place of the question mark (?) in the following series? 800, 700, 600, 500, 400, ?

(A) 399
(B) 789
(C) 656
(D) 300

Q.14

In a certain code language, 'LIVE' is coded as '2648' and 'EAST' is coded as '3215'. What is the code for 'E' in that language?

(A) 8
(B) 3
(C) 2
(D) 5

Q.15

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts,decide which of the given conclusions logically follow(s) from the statements. Statements: All gardens are museums. All museums are parks. Conclusions: (I) Some museums are gardens. (II) All gardens are parks.

(A) Neither conclusion I nor II follows
(B) Both conclusions I and II follow
(C) Only conclusion I follows
(D) Only conclusion II follows

Q.16

Four number triads have been given, out of which three are alike in some manner and one is different. Select the number triad that is different from the rest. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) (8, 16, 256)
(B) (9, 18, 320)
(C) (3, 6, 36)
(D) (5, 10, 100)

Q.17

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: Some desktops are monitors. No monitor is a television. Conclusions: (I) Some desktops are televisions. (II) Some monitors are desktops.

(A) Only conclusion (I) follows.
(B) Only conclusion (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Neither conclusion (I) nor (II) follows.

Q.18

Sunita, Lokesh, Babita, Isha, Akram, Alok and Kiran are sitting around a circular table facing the centre (but not necessarily in the same order). Sunita is an immediate neighbour of Babita. Lokesh sits 2nd to the right of Babita. Isha sits 4th to the right of Akram. Alok sits 3rd to the right of Lokesh. Isha sits 2nd to the left of Alok. Who sits third to the right of Babita?

(A) Alok
(B) Isha
(C) Kiran
(D) Lokesh

Q.19

What should come in place of the question mark (?) in the following series based on the English alphabetical order? ISC, GPF, EMI, CJL, ?

(A) AGP
(B) BGP
(C) AGO
(D) BGO

General Awareness 25 questions

Q.1

iGOT Karmayogi is an integrated online training platform to provide continuous learning opportunities to _____ in order to upgrade their skills.

(A) Journalists
(B) Individual Civil Service Officials
(C) Lawyers
(D) Unorganised Workers

Q.2

'Dashain' and 'Tihar' are popular festivals of which neighbouring country of India?

(A) Sri Lanka
(B) Bangladesh
(C) Nepal
(D) Pakistan

Q.3

Which of the following is an example of a private sector Industry?

(A) Steel Authority of India Limited
(B) Life Insurance Corporation
(C) Bharat Heavy Electricals Limited
(D) Tata Iron and Steel Company

Q.4
Which of the following countries won the maximum number of T20 Women's Cricket World Cup titles?

(A) West Indies
(B) England
(C) India
(D) Australia

Q.5
All natural earthquakes take place in the:

(A) Asthenosphere
(B) Lithosphere
(C) Biosphere
(D) Hydrosphere

Q.6
In February 2024, who was sworn in as the first woman Chief Justice of the Uttarakhand High Court?

(A) Ranjana Desai
(B) Ritu Bahri
(C) Gyan Sudha Mishra
(D) Sujata Manohar

Q.7
Which of the following Acts provided for a system of tribunals and a ban on strikes?

(A) Government of India Act, 1912
(B) The Trade Disputes Act, 1929
(C) Age of Consent Act, 1891
(D) Indian Partnership Act, 1932

Q.8
Which of the following is NOT the principle laid down by the OECD about budgetary governance?

(A) Design the capital budgeting framework to meet national development needs.
(B) Alignment of budgets with medium-term strategic priorities of the government.
(C) It does not allow the citizen to participate in budgetary debates.
(D) The execution of the budget must be proactively planned, managed, and monitored.

Q.9
Epic Poem, Savitri, described by its author as ‘A Legend and a Symbol’, was written by _____.

(A) Aurobindo Ghosh
(B) Baba Saheb Ambedkar
(C) Bal Gangadhar Tilak
(D) Mahatma Phule

Q.10
Which chronicler of the medieval age recognised that Raziyya was more able and qualified than all her brothers?

(A) Minhaj-i-Siraj
(B) Taj-ud-din
(C) Muhammad Junaidi
(D) Fakhrul-Mulk-Isami

Q.11
Which of the following fields is NOT covered by the National Eminence Award?

(A) Literature
(B) Music of all forms and origin
(C) Theatre
(D) Dance

Q.12
Which specific chemical can be applied to achieve staining of chromosomes?

(A) Ethanol
(B) Methanol
(C) Hydrochloric acid
(D) Acetocarmine

Q.13
Which track and field event has a water jump?

(A) Hurdles running
(B) Steeplechase
(C) Decathlon
(D) Cross country running

Q.14
According to the Census of India 2011, what is the population density of India?

(A) 382/km²
(B) 388/km²
(C) 375/km²
(D) 360/km²

Q.15
Which of the following Indian tribal dances is prevalent in the Araku Valley near Visakhapatnam?

(A) Bardo Chham
(B) Dhimsa
(C) Bagurumba
(D) Jhumur

Q.16
Which Article of the Indian Constitution has provisions for the appointment of High Court Judges?

(A) Article 119
(B) Article 252
(C) Article 217
(D) Article 202

Q.17
In which of the following years did Alexander invade Punjab and in which of the following years did he die?

(A) 326 BC and 323 BC
(B) 323 BC and 328 BC
(C) 320 BC and 323 BC
(D) 326 BC and 327 BC

Q.18
Carnatic singer Bombay Jayashri has been selected for which music award in 2023?

(A) Sangeetha Kalasikhamani
(B) Sangeetha Choodamani
(C) Sangita Kalanidhi
(D) Sangeetha Kalasarathy

Q.19
JJ Thomson is known for developing which model in 1898?

(A) Gold foil model
(B) Oil drop model
(C) Double-slit model
(D) Plum pudding model

Q.20
Right of a prisoner to speedy trial will be categorised in which of the following Fundamental Rights?

(A) Right to Equality
(B) Right to Life
(C) Right against Exploitation
(D) Right to Constitutional Remedies

Q.21
The Zaskar and the Nubra are tributaries of which river?

(A) Ganga
(B) Kaveri
(C) Mahanadi
(D) Indus

Q.22
In which of the following states is Aoleang Monyü celebrated, which is an ancient spring festival celebrated by all Konyaks in the month of April?

(A) Telangana
(B) Bihar
(C) Odisha
(D) Nagaland

Q.23
'Once a Dancer' is the autobiography of _____.

(A) Sudha Chandran
(B) Audrey Hepburn
(C) Yamini Krishnamurthy
(D) Allegra Kent

Q.24

If after being bitten by a mosquito, the person is detected as being malaria positive, which of the following will hold true? A. The person did not have a balanced diet. B. The mosquito was carrying plasmodium. C. The mosquito was suffering from malaria. D. The person was not vaccinated as a child.

(A) C
(B) D
(C) A
(D) B

Q.25

Who excavated the Chira Tope stupa at Taxila?

(A) James Mill
(B) James Princip
(C) John Marshall
(D) Alexander Cunningham

English Language and Comprehension 24 questions

Q.1

Select the most appropriate option to fill in the blank. The teacher encouraged students to cultivate a _____ attitude towards learning, embracing challenges as opportunities for growth.

(A) redundant
(B) complacent
(C) stagnant
(D) resilient

Q.2

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. When they left for Mexico, their house was in “good” condition than it is now.

(A) best
(B) better
(C) excellent
(D) more good

Q.3

In the following sentence the underlined part contains an error. Below are given alternatives to the underlined part. Select the correct alternative. Kanha National Park in MP is the “dense” forest in India.

(A) most dense
(B) densest
(C) denser
(D) more denser

Q.4

Select the most appropriate adjective to fill in the blank. The Nile is the _____ river in the world.

(A) long
(B) most long
(C) longer
(D) longest

Q.5

Select the most appropriate option that can substitute the underlined segment in the given sentence. The government is developing a new test “in the recruitment and training of teachers” in the upcoming session.

(A) of the recruitment and training of teachers
(B) for the recruitment and training of teachers
(C) in recruitment and training on teachers
(D) for recruitment or training of teachers

Q.6

Select the most appropriate phrase that can substitute the underlined segment in the given sentence. The question paper was “very easy to complete”.

(A) the last straw
(B) a snowball effect
(C) fair and square
(D) a piece of cake

Q.7

Select the most appropriate meaning of the underlined word in the context of the following sentence. The “palm” is a native of Malaysia. We can find them in groves.

(A) A type of soil
(B) Inner surface of the hand
(C) A pamphlet
(D) A type of tree

Q.8

Select the option that rectifies the spelling of the underlined word in the given sentence. How can one justify sending sound waves into space when they cannot travel through a “vacum”?

(A) vaccum
(B) vaccuum
(C) vacume
(D) vacuum

Q.9

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution required’. The car, with all “its features, were meticulously” designed.

(A) No substitution required
(B) its features, have meticulously
(C) it's features, were meticulously
(D) its features, was meticulously

Q.10

Select the appropriate synonym of the underlined word. The bride has an “exquisite” face.

(A) Ugly
(B) Average
(C) Plain
(D) Beautiful

Q.11

Select the most appropriate synonym of the given word. Preserve

(A) Maintain
(B) Ignore
(C) Complain
(D) Consume

Q.12

Select the most appropriate idiom to fill in the blank. We hear the judges from their _____ that only CNG buses should run in Delhi.

(A) ivory tower
(B) square meal
(C) melting pot
(D) look sharp

Q.13

Identify the sentence with an error in the use of simple present tense.

(A) He go to school every day.
(B) You do yoga every morning.
(C) She walks every morning.
(D) We brush our teeth daily.

Q.14

Select the most appropriate meaning of the underlined word. He took the “right” decision at that time.

(A) A particular direction
(B) Something which is correct
(C) Someone’s liberty and freedom
(D) Something which is invalid

Q.16

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. She received / an award / for her work / in a engineering field.

(A) for her work
(B) an award
(C) She received
(D) in a engineering field.

Q.17

Select the most appropriate synonym of the underlined word in the given sentence. Her fear of public speaking was a major “hurdle” in her career.

(A) Support
(B) Stimulus
(C) Encouragement
(D) Obstacle

Q.18

In the given sentence, one of the four underlined words is incorrectly spelt. Identify the INCORRECTLY spelt word. The scientific “community” gave “credance” to the new research findings after “rigorous” testing and “validation”.

(A) credance
(B) validation
(C) community
(D) rigorous

Q.19

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. There is only one way / to have a happier marred life / and I will learn it soon.

(A) to have a happier marred life
(B) No error
(C) and I will learn it soon.
(D) There is only one way

Q.20

Select the INCORRECTLY spelt word in the following sentence. The restaurant's specialtiy is seafood.

(A) seafood
(B) restaurant's
(C) The
(D) specialtiy

Q.21

In the following passage, some words have been deleted. Fill in the blanks with the help of the alternatives given. Select the most appropriate option for each number. How environmental problems ____1_____ different groups is a function of social inequality. Social status and power determine the _____2____ to which people can insulate themselves from environmental crises or overcome them. In some cases, their ‘solutions’ may actually worsen environmental disparities. In Kutch, Gujarat, where water is scarce, richer farmers have ____3____ in deep bore tubewells to tap groundwater to irrigate their fields and grow cash crops. When the rains fail, the earthen wells of the poorer villagers run dry and they do not even have water to drink. At such times, the moist green fields of the rich farmers seem to mock them. Certain environmental concerns sometimes appear to be universal concerns, not particular to ____4_____ social groups. For instance, _____5_____ air pollution or protecting biodiversity seems to be in the public interest. A sociological analysis shows, however, that how public priorities are set and how they are pursued may not be universally beneficial.

Select the most appropriate option to fill in blank number 1.

(A) Partial
(B) Adopt
(C) Effect
(D) Affect

Q.22

In the following passage, some words have been deleted. Fill in the blanks with the help of the alternatives given. Select the most appropriate option for each number. How environmental problems ____1_____ different groups is a function of social inequality. Social status and power determine the _____2____ to which people can insulate themselves from environmental crises or overcome them. In some cases, their 'solutions' may actually worsen environmental disparities. In Kutch, Gujarat, where water is scarce, richer farmers have ____3____ in deep bore tubewells to tap groundwater to irrigate their fields and grow cash crops. When the rains fail, the earthen wells of the poorer villagers run dry and they do not even have water to drink. At such times, the moist green fields of the rich farmers seem to mock them. Certain environmental concerns sometimes appear to be universal concerns, not particular to ____4_____ social groups. For instance, _____5_____ air pollution or protecting biodiversity seems to be in the public interest. A sociological analysis shows, however, that how public priorities are set and how they are pursued may not be universally beneficial.

Select the most appropriate option to fill in blank number 2.

(A) Extent
(B) Scope
(C) Various
(D) Expanse

Q.23

In the following passage, some words have been deleted. Fill in the blanks with the help of the alternatives given. Select the most appropriate option for each number. How environmental problems ____1_____ different groups is a function of social inequality. Social status and power determine the _____2____ to which people can insulate themselves from environmental crises or overcome them. In some cases, their 'solutions' may actually worsen environmental disparities. In Kutch, Gujarat, where water is scarce, richer farmers have ____3____ in deep bore tubewells to tap groundwater to irrigate their fields and grow cash crops. When the rains fail, the earthen wells of the poorer villagers run dry and they do not even have water to drink. At such times, the moist green fields of the rich farmers seem to mock them. Certain environmental concerns sometimes appear to be universal concerns, not particular to ____4_____ social groups. For instance, _____5_____ air pollution or protecting biodiversity seems to be in the public interest. A sociological analysis shows, however, that how public priorities are set and how they are pursued may not be universally beneficial.

Select the most appropriate option to fill in blank number 3.

(A) Invested
(B) Licensed
(C) Provided
(D) Capitalised

Q.24

In the following passage, some words have been deleted. Fill in the blanks with the help of the alternatives given. Select the most appropriate option for each number. How environmental problems ____1_____ different groups is a function of social inequality. Social status and power determine the _____2____ to which people can insulate themselves from environmental crises or overcome them. In some cases, their 'solutions' may actually worsen environmental disparities. In Kutch, Gujarat, where water is scarce, richer farmers have ____3____ in deep bore tubewells to tap groundwater to irrigate their fields and grow cash crops. When the rains fail, the earthen wells of the poorer villagers run dry and they do not even have water to drink. At such times, the moist green fields of the rich farmers seem to mock them. Certain environmental concerns sometimes appear to be universal concerns, not particular to ____4_____ social groups. For instance, _____5_____ air pollution or protecting biodiversity seems to be in the public interest. A sociological analysis shows, however, that how public priorities are set and how they are pursued may not be universally beneficial.

Select the most appropriate option to fill in blank number 4.

(A) Indefinitely
(B) Specific
(C) Exactly
(D) Restriction

Q.25

In the following passage, some words have been deleted. Fill in the blanks with the help of the alternatives given. Select the most appropriate option for each number. How environmental problems ____1_____ different groups is a function of social inequality. Social status and power determine the _____2____ to which people can insulate themselves from environmental crises or overcome them. In some cases, their 'solutions' may actually worsen environmental disparities. In Kutch, Gujarat, where water is scarce, richer farmers have ____3____ in deep bore tubewells to tap groundwater to irrigate their fields and grow cash crops. When the rains fail, the earthen wells of the poorer villagers run dry and they do not even have water to drink. At such times, the moist green fields of the rich farmers seem to mock them. Certain environmental concerns sometimes appear to be universal concerns, not particular to ____4_____ social groups. For instance, _____5_____ air pollution or protecting biodiversity seems to be in the public interest. A sociological analysis shows, however, that how public priorities are set and how they are pursued may not be universally beneficial.

Select the most appropriate option to fill in blank number 5.

(A) Manipulating

(B) Increasing

(C) Abdicating

(D) Reducing

Answer key

Numerical and Mathematical Ability

Q.2 **(B)**	Q.3 **(C)**	Q.5 **(C)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(D)**
Q.12 **(D)**	Q.13 **(D)**	Q.15 **(A)**	Q.16 **(B)**	Q.18 **(A)**	Q.19 **(B)**
Q.20 **(A)**					

Reasoning Ability and Problem Solving

Q.2 **(A)**	Q.3 **(C)**	Q.6 **(A)**	Q.7 **(A)**	Q.10 **(B)**	Q.12 **(C)**
Q.13 **(D)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(B)**
Q.19 **(C)**					

General Awareness

Q.1 **(B)**	Q.2 **(C)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(B)**	Q.8 **(C)**	Q.9 **(A)**	Q.10 **(A)**	Q.11 **(A)**	Q.12 **(D)**
Q.13 **(B)**	Q.14 **(A)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(C)**
Q.19 **(D)**	Q.20 **(B)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(B)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(D)**
Q.7 **(D)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(A)**
Q.13 **(A)**	Q.14 **(B)**	Q.16 **(D)**	Q.17 **(D)**	Q.18 **(A)**	Q.19 **(A)**
Q.20 **(D)**	Q.21 **(D)**	Q.22 **(A)**	Q.23 **(A)**	Q.24 **(B)**	Q.25 **(D)**

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (B) 60.725

The 41 numbers total 41 × 62 = 2,542. The first 18 total 18 × 65.5 = 1,179 and the last 24 total 24 × 61.5 = 1,476; together they count the 18th number twice, so it is 1,179 + 1,476 − 2,542 = 113. The other 40 average (2,542 − 113)/40 = 60.725.

Q.3 · Answer (C) Decrease by 4.2%

On an income of 100 she spends 76 and saves 24. Now her income is 118 and her spending 76 × 1.25 = 95, so she saves 23: a fall of 1/24 ≈ 4.2%.

Q.5 · Answer (C) 3 : 2

With a lid the cone's total surface area is $\pi r(l + r)$. Setting it equal to πR^2 with $R = (l + r)/2$ gives $(l + r)^2/4 = r(l + r)$, so $l + r = 4r$: $l = 3r$ and $R = 2r$. $l : R = 3 : 2$.

Q.9 · Answer (A) ₹200

28% then 25% leaves 0.72 × 0.75 = 0.54 of the price (46% off); 10% then 5% leaves 0.9 × 0.95 = 0.855 (14.5% off). The difference, 31.5% of the marked price, is ₹63, so the price is ₹200.

Q.10 · Answer (C) 1800

$120 = 2^3 \times 3 \times 5$, $200 = 2^3 \times 5^2$ and $225 = 3^2 \times 5^2$. LCM $= 2^3 \times 3^2 \times 5^2 = 1800$.

Q.11 · Answer (D) 4.5 km/h and 2.5 km/h

Downstream speed = 21/3 = 7 km/h and upstream speed = 14/7 = 2 km/h. Boat = (7 + 2)/2 = 4.5 km/h and river = (7 − 2)/2 = 2.5 km/h.

Q.12 · Answer (D) 238.5

At the same speed, (L + 900)/110 = (L + 486)/70. So 70L + 63,000 = 110L + 53,460, 40L = 9,540 and L = 238.5 m.

Q.13 · Answer (D) 280

SI = 3500 × 4 × 2/100 = ₹280.

Q.15 · Answer (A) 27

The bowl holds $(2/3)\pi \times 27^3 = 13{,}122\pi$ cm³ and each bottle (radius 6 cm) $\pi \times 6^2 \times 13.5 = 486\pi$ cm³. 13,122 ÷ 486 = 27.

Q.16 · Answer (B) 100 kg

The total rises by 11 × 4.5 = 49.5 kg, so the new person weighs 50.5 + 49.5 = 100 kg.

Q.18 · Answer (A) 110

The five total 5 × 62 = 310; the first two total 96 and the last two 104, so the third is 310 − 96 − 104 = 110.

Q.19 · Answer (B) 1500

P = SI × 100/(R × T) = 240 × 100/(8 × 2) = ₹1500.

Q.20 · Answer (A) 2

(5 + x)/(19 + x) = (7 + x)/(25 + x) gives 125 + 30x = 133 + 26x, so x = 2. Check: 7 : 21 = 9 : 27.

Reasoning Ability and Problem Solving

Q.2 · Answer (A) 1

Facing outwards, each person's right is clockwise. The clues fit only one order, clockwise: D, F, E, A, C, G, B. Counting from G to the right comes B and then D, so one person sits between them.

Q.3 · Answer (C) MSKG

Each letter moves 3 places back: R → O, X → U, P → M, L → I, and likewise TZRN → QWOK. So PVNJ → MSKG.

Q.6 · Answer (A) 101
Each number is related to its square plus 1: $4^2 + 1 = 17$ and $5^2 + 1 = 26$. So $10^2 + 1 = 101$.

Q.7 · Answer (A) 72
After interchanging the signs: $69 + 24 \times 4 \div 6 - 13 = 69 + 16 - 13 = 72$.

Q.10 · Answer (B) ka
'food' is the only word in both phrases and 'ka' the only code in both, so food = ka.

Q.12 · Answer (C) 7PM5
The digits rise by 2, the first letter moves 2 places forward and the second 2 places back: 4KC2 → 6MA4 and 3MN4 → 5OL6. So 5NO3 → 7PM5.

Q.13 · Answer (D) 300
Each term is 100 less than the one before: 400 − 100 = 300.

Q.14 · Answer (C) 2
E is the only letter in both words and 2 the only digit in both codes, so E = 2.

Q.15 · Answer (B) Both conclusions I and II follow
All gardens are museums, so some museums are gardens (I). All gardens are museums and all museums are parks, so all gardens are parks (II). Both follow.

Q.16 · Answer (B) (9, 18, 320)
In the other triads the second number is twice the first and the third is the square of the second: $16^2 = 256$, $6^2 = 36$, $10^2 = 100$. But $18^2 = 324$, not 320.

Q.17 · Answer (B) Only conclusion (II) follows.
'Some desktops are monitors' means some monitors are desktops (II). The desktops that are monitors cannot be televisions, and nothing says any other desktop is one, so I does not follow.

Q.18 · Answer (B) Isha
Counting places to the right from Babita: Lokesh is 2nd, Alok 5th (3rd to Lokesh's right), Isha 3rd (2nd to Alok's left) and Akram 6th (Isha is 4th to his right). Sunita, Babita's neighbour, takes the 1st place and Kiran the 4th. Third to Babita's right is Isha.

Q.19 · Answer (C) AGO
First letters move 2 back (I, G, E, C, A), second letters 3 back (S, P, M, J, G) and third letters 3 forward (C, F, I, L, O): AGO.

General Awareness

Q.1 · Answer (B) Individual Civil Service Officials
iGOT (Integrated Government Online Training) Karmayogi, under Mission Karmayogi, is the online learning platform for building the capacity of civil servants.

Q.2 · Answer (C) Nepal
Dashain and Tihar (the Nepali festival of lights) are the main Hindu festivals of Nepal.

Q.3 · Answer (D) Tata Iron and Steel Company
TISCO (now Tata Steel), a Tata Group company, is privately owned. SAIL, LIC and BHEL are owned by the Government of India.

Q.4 · Answer (D) Australia
By 2024 Australia had won the Women's T20 World Cup six times (2010, 2012, 2014, 2018, 2020 and 2023), more than any other side.

Q.5 · Answer (B) Lithosphere
All natural earthquakes take place in the lithosphere, the rigid outer layer of the Earth made up of the crust and the uppermost mantle.

Q.6 · Answer (B) Ritu Bahri
Justice Ritu Bahri, earlier of the Punjab and Haryana High Court, became the first woman Chief Justice of the Uttarakhand High Court.

Q.7 · Answer (B) The Trade Disputes Act, 1929
The Trade Disputes Act, 1929 set up courts of inquiry and boards of conciliation to settle industrial disputes and made strikes in public utility services illegal without advance notice.

Q.8 · Answer (C) It does not allow the citizen to participate in budgetary debates.
The OECD's 2015 Recommendation on Budgetary Governance calls for an inclusive, participative and realistic debate on budget choices, so shutting citizens out of budget debates goes against it. The other three are among its principles.

Q.9 · Answer (A) Aurobindo Ghosh
Sri Aurobindo (Aurobindo Ghosh) wrote 'Savitri: A Legend and a Symbol', an epic poem in blank verse on the story of Savitri and Satyavan.

Q.10 · Answer (A) Minhaj-i-Siraj
Minhaj-i-Siraj, a chronicler of the Delhi Sultanate, recognised Raziyya's ability, though he was uneasy at having a queen as ruler.

Q.11 · Answer (A) Literature
The National Eminence Award of the Sri Shanmukhananda Fine Arts and Sangeetha Sabha, Mumbai, set up in 2002, honours music of all forms and origin, dance, theatre and fine arts administrators. Literature is not one of its fields.

Q.12 · Answer (D) Acetocarmine
Acetocarmine stains chromosomes, which is why it is used to study cell division, for example in onion root tips.

Q.13 · Answer (B) Steeplechase
The steeplechase (usually over 3000 m) has barriers and a water jump on each full lap.

Q.14 · Answer (A) 382/km²
The 2011 Census put India's population density at 382 persons per km².

Q.15 · Answer (B) Dhimsa
Dhimsa is a dance of the tribal communities of the Araku Valley in Andhra Pradesh; Bardo Chham belongs to Arunachal Pradesh and Bagurumba to the Bodos of Assam.

Q.16 · Answer (C) Article 217
Article 217 deals with the appointment and conditions of office of a Judge of a High Court.

Q.17 · Answer (A) 326 BC and 323 BC
Alexander crossed into Punjab in 326 BC, defeating Porus at the Hydaspes (Jhelum), and died at Babylon in 323 BC.

Q.18 · Answer (C) Sangita Kalanidhi
The Music Academy, Madras chose Bombay Jayashri for its Sangita Kalanidhi, the highest honour in Carnatic music, in 2023.

Q.19 · Answer (D) Plum pudding model
Thomson pictured the atom as a sphere of positive charge with electrons embedded in it, like plums in a pudding. The gold foil experiment was Rutherford's and the oil drop experiment Millikan's.

Q.20 · Answer (B) Right to Life
The Supreme Court has held the right to a speedy trial to be part of the right to life and personal liberty under Article 21 (Hussainara Khatoon v. State of Bihar).

Q.21 · Answer (D) Indus
The Zaskar and the Nubra are tributaries of the Indus, as are the Shyok, the Gilgit, the Hunza and the Dras.

Q.22 · Answer (D) Nagaland
Aoleang Monyü, the spring festival of the Konyak Nagas, is celebrated in Nagaland's Mon district in the first week of April.

Q.23 · Answer (D) Allegra Kent
'Once a Dancer' (1997) is the autobiography of Allegra Kent, a ballerina of the New York City Ballet.

Q.24 · Answer (D) B
Malaria is caused by the Plasmodium parasite, which the female Anopheles mosquito carries, so B must hold. The mosquito does not itself suffer from the disease, and a positive test says nothing about the person's diet or childhood vaccinations.

Q.25 · Answer (C) John Marshall
Sir John Marshall, Director-General of the Archaeological Survey of India, excavated Taxila, including the Dharmarajika stupa known as Chir Tope.

English Language and Comprehension

Q.1 · Answer (D) resilient
'Resilient' means able to recover from difficulties, which fits treating challenges as chances to grow; the other options are negative.

Q.2 · Answer (B) better
'Than' calls for the comparative: 'in better condition than it is now'.

Q.3 · Answer (B) densest
'The … forest in India' calls for the superlative, and one-syllable 'dense' takes '-est': 'the densest forest'.

Q.4 · Answer (D) longest
Comparison with every river in the world calls for the superlative: 'the longest river'.

Q.5 · Answer (B) for the recruitment and training of teachers
A test is developed 'for' a purpose: 'a new test for the recruitment and training of teachers'.

Q.6 · Answer (D) a piece of cake
'A piece of cake' is something very easy.

Q.7 · Answer (D) A type of tree
Here 'palm' is the tree: palms grow in groves. The palm of the hand is another meaning.

Q.8 · Answer (D) vacuum
The correct spelling is 'vacuum'.

Q.9 · Answer (D) its features, was meticulously
The subject is 'The car'; 'with all its features' does not make it plural, so 'was meticulously designed'. 'It's' means 'it is'.

Q.10 · Answer (D) Beautiful
'Exquisite' means extremely beautiful.

Q.11 · Answer (A) Maintain
To 'preserve' something is to keep it as it is: to 'maintain' it.

Q.12 · Answer (A) ivory tower
An 'ivory tower' is a place of privileged seclusion, cut off from everyday realities, from which such a pronouncement is made.

Q.13 · Answer (A) He go to school every day.
With 'he' the simple present takes the third-person singular form: 'He goes to school every day.'

Q.14 · Answer (B) Something which is correct
Here 'right' means correct: he made the correct decision.

Q.16 · Answer (D) in a engineering field.
'An' comes before a vowel sound: 'in an engineering field'.

Q.17 · Answer (D) Obstacle
A 'hurdle' is a difficulty to be overcome: an 'obstacle'.

Q.18 · Answer (A) credance
The correct spelling is 'credence' (belief or acceptance).

Q.19 · Answer (A) to have a happier marred life
'Marred' means spoiled; the sense calls for 'married': 'a happier married life'.

Q.20 · Answer (D) specialtiy
The correct spelling is 'specialty' (or 'speciality').

Q.21 · Answer (D) Affect
How environmental problems 'affect' different groups: 'affect' is the verb, 'effect' usually a noun.

Q.22 · Answer (A) Extent
Social status and power determine 'the extent to which' people can insulate themselves.

Q.23 · Answer (A) Invested
Richer farmers have 'invested in' deep bore tubewells.

Q.24 · Answer (B) Specific
Concerns that seem universal, not particular to 'specific' social groups.

Q.25 · Answer (D) Reducing
'Reducing' air pollution, like protecting biodiversity, seems to be in the public interest.

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